PRODUCT OVERVIEW

Pharma Scope

An autonomous investigator for public pharma and healthcare information.

The problem and intended outcome

Financial stories are spread across reports, filings, regulatory notices and industry sources. Pharma Scope searches for those pieces, calculates relationships, proposes explanations and tests alternatives. The goal is a non-obvious, auditable financial finding; unsupported interpretations are withheld.

How the pipeline works

1 Identify Resolve the company from fetched passages; check its name/symbol and Healthcare classification against a dated official NIFTY 50 snapshot.

2 Plan Choose business-model, geographic or regulatory research angles and targeted searches without a preset company conclusion.

3 Collect Search public pages, follow full-disclosure links, and archive readable PDFs/HTML with original URLs, timestamps and hashes.

4 Validate Extract structured facts. Require a separate fiscal-year column-header quote for multi-value rows, then check quotations, values, periods, entity and reporting basis.

5 Calculate Python calculates comparable growth, shares, margins, differences or CAGR. Hypotheses cite real fact and calculation IDs.

6 Challenge Gather supporting and opposing evidence; test currency, acquisitions, reporting changes and other explanations. Missing evidence stays unresolved.

7 Audit Draft the required output, audit claim support, and apply publication gates. Display a supported/unresolved finding or the evidence gap.

What a qualifying finding contains

Discovery; three to five significance bullets; linked evidence; timeline; necessary calculations; strongest counterargument; measurable future falsification condition; and a reasoned Strong/Moderate/Weak rating. A rating is not a probability of truth.

Operating model

A local browser dashboard talks to Python. LangGraph coordinates research, Gemini interprets supplied evidence, DDGS searches, PyMuPDF/Beautiful Soup read documents, and SQLite stores checkpoints. Search and model calls use the internet. Keys stay in the backend.

VALIDATION AND LIMITATIONS

What the live trials show

Eight-company evaluation, 25 Oct 2026 IST. Same Wider budgets; two workers. A general identity guard and comparable-pair guidance were added after the first six trials. Earlier evidence was revalidated under the current policy.

Company	Cycle (s)	Financial inputs	Published findings
Dr. Reddy's	121.7	0	0
Cipla	162.8	4	0
Lupin	17.3	0	0
Sun Pharma	197.1	27	0
Zydus Lifesciences	163.8	0	0
Apollo Hospitals	80.1	0	0
Max Healthcare	40.1	0	0
Divi's	98.3	4	0

Real walkthrough: Sun Pharma

No published discovery. The run retained 27 financial inputs. Its cited consolidated table reports FY2025/FY2026 revenue from operations of INR 525,784.4/584,620.4 million, a checked increase of 11.19%. The original page and column order were visually inspected. The proposed structural-growth explanation remained unresolved: organic growth, currency and acquisition effects were not established.

Source: Sun Pharma annual report 2025-26, physical page 242 (printed page 240). This is an auditable observation, not a demonstrated non-obvious discovery.

Bounded 270s cycle

Both modes use 270s of research. Standard: up to 8 searches / 12 source requests / 20 model calls / 6 pages per source. Wider: 10 / 16 / 24 / 8. Wider changes coverage, not time. A supervisor stops a stalled worker at 280s; saving and display can add overhead. No result or accuracy guarantee follows from these limits.

Current gaps and safeguards

No supported or publication-qualified unresolved discovery emerged from these trials. The Apollo Hospitals identity mismatch was excluded. Search rate limits and unreadable sources reduced coverage. Scanned-report OCR, first-mainstream-coverage timing, complete five-year drivers and broad independent corroboration remain incomplete. Snapshot eligibility is not live/historical membership. No investment recommendation is produced.

Read RUNBOOK.md for operation, APPENDIX.md for validation rules, EVALUATION.md for all outcomes, and REAL_CASE.md for source passages and the unresolved test. Software tests do not prove financial truth.